

# Project Report Format

## CREDIT CARD APPROVAL PREDICTION

### 1. INTRODUCTION

#### 1.1 Project Overview

The **Credit Card Approval Prediction** system is a machine learning-based web application that predicts whether a customer's credit card application is likely to be approved or rejected based on financial and personal information. Banks receive thousands of applications every day, making manual verification slow and error-prone. This project automates the approval process using machine learning algorithms, enabling faster, more accurate, and unbiased decisions.

The application is developed using **Python, Flask, Scikit-learn, HTML, CSS, and JavaScript**, with trained machine learning models integrated into a user-friendly web interface.

#### 1.2 Purpose

The main purpose of this project is to develop an intelligent system that assists financial institutions in evaluating credit card applications automatically. The system reduces manual effort, improves prediction accuracy, minimizes processing time, and supports data-driven decision-making.

## 2. IDEATION PHASE

### 2.1 Problem Statement

Banks process a large number of credit card applications daily. Manual verification is time-consuming, expensive, and prone to human error. There is a need for an automated system capable of accurately predicting credit card approval decisions using applicant information and machine learning techniques.

## 2.2 Empathy Map Canvas

### Think & Feel

- Wants quick approval
- Concerned about eligibility
- Expects fair evaluation
- Wants transparency

### See

- Lengthy application process
- Different bank policies
- Multiple documentation requirements

### Hear

- Advice from bank staff
- Customer reviews
- Financial recommendations

### Say & Do

- Submit application
- Upload required documents
- Wait for approval
- Compare different banks

### Pain

- Long waiting time
- Application rejection
- Unclear eligibility criteria
- Manual verification delays

### Gain

- Instant prediction
- Faster processing
- Accurate evaluation
- Better customer experience

## 2.3 Brainstorming

The team discussed several ideas before finalizing the project:

- Collect credit card applicant dataset
- Perform exploratory data analysis
- Clean and preprocess data
- Train multiple machine learning models
- Compare model accuracy
- Develop Flask web application
- Display prediction results instantly
- Deploy the application online

## 3. REQUIREMENT ANALYSIS

### 3.1 Customer Journey Map

1. Customer visits the application.
2. Enters personal and financial information.
3. Clicks **Predict**.
4. Data is validated.
5. Machine learning model processes inputs.
6. Prediction result is displayed.
7. Customer reviews approval status.

### 3.2 Solution Requirements

#### Functional Requirements

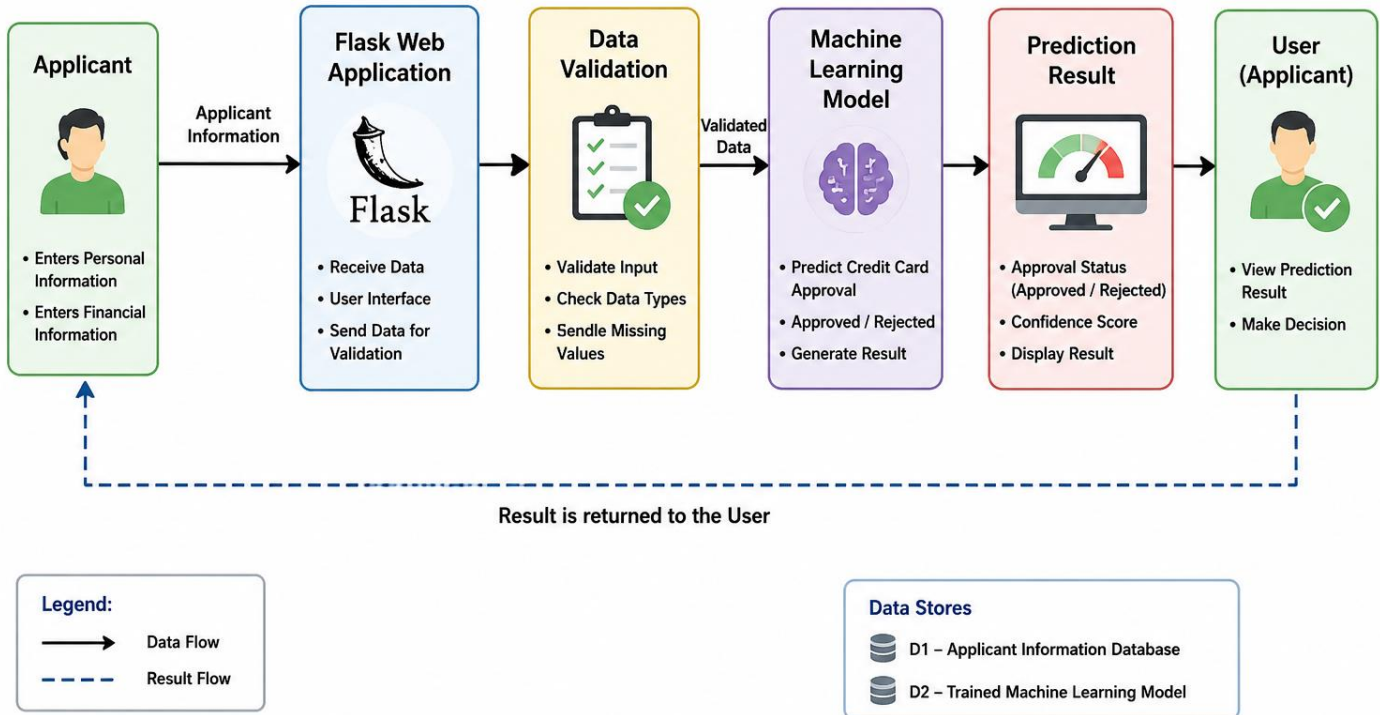
- Input applicant information
- Validate user inputs
- Predict approval status
- Display prediction confidence
- Show approval/rejection result

#### Non-Functional Requirements

- Fast response time
- High prediction accuracy
- Secure data processing
- Responsive user interface
- Easy deployment

### 3.3 Data Flow Diagram

#### Data Flow Diagram – Credit Card Approval Prediction



### 3.4 Technology Stack

| Category             | Technology            |
|----------------------|-----------------------|
| Programming Language | Python                |
| Machine Learning     | Scikit-learn          |
| Web Framework        | Flask                 |
| Frontend             | HTML, CSS, JavaScript |
| Data Analysis        | Pandas, NumPy         |
| Visualization        | Matplotlib, Seaborn   |
| Model Storage        | Joblib                |
| IDE                  | Visual Studio Code    |
| Version Control      | Git & GitHub          |

# 4. PROJECT DESIGN

## 4.1 Problem Solution Fit

The proposed solution automates the credit card approval process by replacing manual verification with machine learning prediction, reducing time and improving decision accuracy.

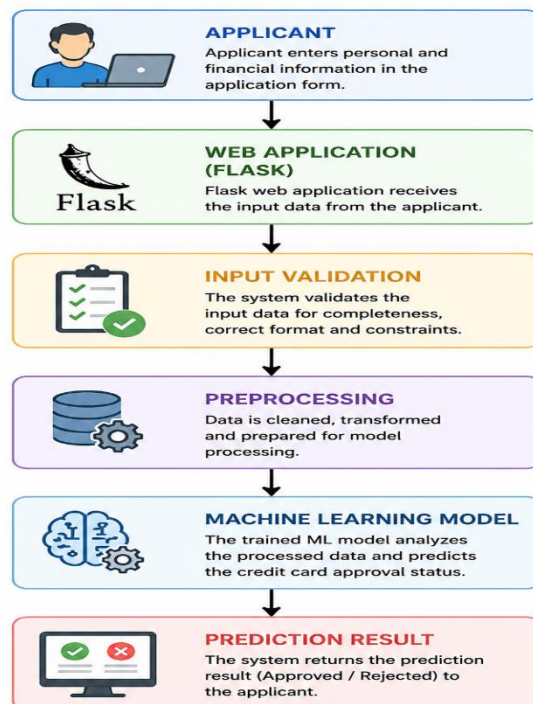
## 4.2 Proposed Solution

- Collect applicant information
- Preprocess data
- Train multiple ML models
- Select the best-performing model
- Deploy using Flask
- Display prediction instantly

## 4.3 Solution Architecture

### SOLUTION ARCHITECTURE

Credit Card Approval Prediction System



# 5. PROJECT PLANNING & SCHEDULING

## Sprint 1

- Project Planning
- Dataset Collection
- ER Diagram

## Sprint 2

- Data Analysis
- Data Visualization
- Descriptive Statistics

## Sprint 3

- Data Cleaning
- Feature Engineering
- Handling Missing Values

## Sprint 4

- Logistic Regression
- Random Forest
- Decision Tree
- Model Evaluation

## Sprint 5

- Flask Development
- Testing
- Deployment
- Documentation

# 6. FUNCTIONAL AND PERFORMANCE TESTING

## 6.1 Functional Testing

- Input validation
- Prediction generation
  
- Model loading
- Web interface testing
- Accuracy verification

## Performance Testing

- Fast prediction response
- Multiple user requests
- Stable model performance
- Successful deployment

# 7. RESULTS

The application successfully predicts whether a credit card application is **Approved** or **Rejected** based on the applicant's financial profile.

The Random Forest model achieved the highest prediction accuracy among all trained models and was selected for deployment.

# 8. ADVANTAGES & DISADVANTAGES

## Advantages

- Fast prediction
- High accuracy
- Reduces manual effort
- Easy to use
- Supports better decision-making
- Web-based application

## Disadvantages

- Depends on data quality
- Requires retraining with new data
- Cannot replace final human approval
- Prediction accuracy depends on training dataset

## 9. CONCLUSION

The **Credit Card Approval Prediction** project successfully demonstrates how machine learning can automate the credit card approval process. The developed Flask application provides quick, accurate, and reliable predictions while minimizing manual effort. The project improves operational efficiency and supports financial institutions in making consistent credit approval decisions.

## 10. FUTURE SCOPE

- Deep Learning-based prediction
- Explainable AI (XAI)
- Cloud deployment
- Mobile application
- Real-time bank database integration
- Applicant authentication
- Dashboard for bank officials
- Continuous model retraining

## 11. APPENDIX

### Source Code

Python, Flask, HTML, CSS, JavaScript

### Dataset

Credit Card Approval Dataset (UCI/Kaggle)

### GitHub Repository

Your GitHub repository link

### Project Demo

Your deployed application link (PythonAnywhere/Render)

This content is suitable for directly filling the sections in your final project report template.